

BVAR vs. Information Criteria: A Monte Carlo Simulation Study

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June 18, 2026

Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Keyword 1, Keyword 2, Keyword 3

JEL Codes: C15, C32

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1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

2 Methodology

2.1 Data Generating Process

Explain the true DGP based on Kilian and Murphy (2014) or other literature.

2.2 Optimal Prior Selection via Marginal Data Density

Describe how the shrinkage parameter (τ) is treated as an optimization problem rather than a fixed heuristic.

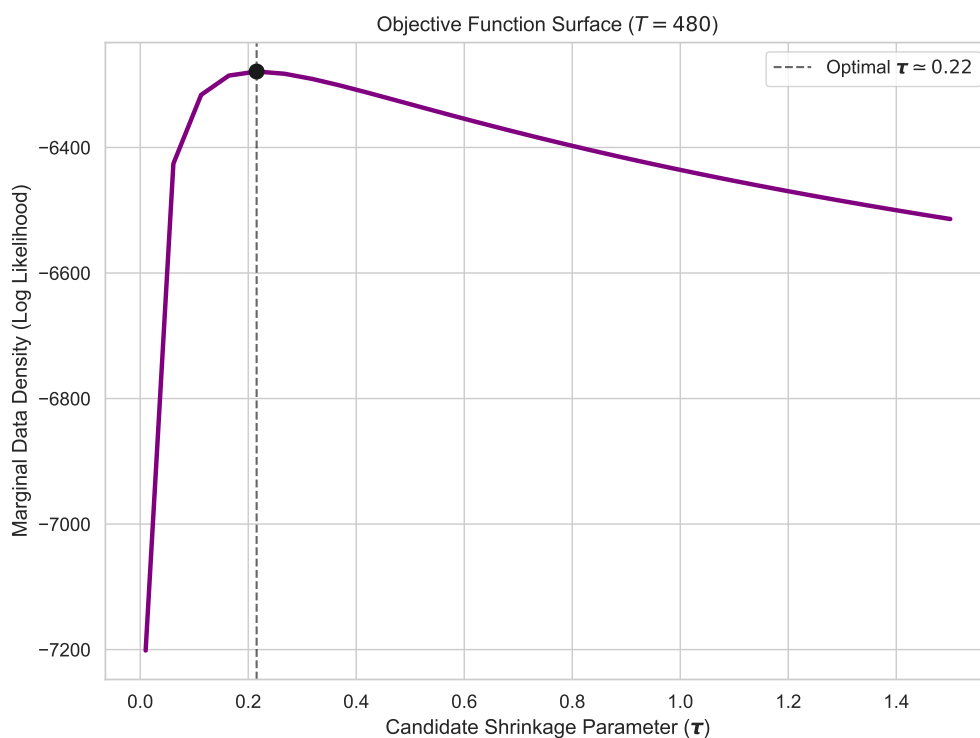


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3 Simulation Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

3.1 Behavior of Standard Information Criteria

Before evaluating the Bayesian estimators, we examine the baseline lag detection performance of the classical Information Criteria.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator	AIC	BIC	HQC
	T			
4	96	12.00	1.50	2.00
	144	2.50	1.00	2.00
	240	3.50	2.00	2.00
	480	3.50	2.00	2.00
6	96	12.00	1.50	2.00
	144	2.50	2.00	2.00
	240	3.00	2.00	2.00
	480	6.00	2.50	2.50
8	96	12.00	1.00	2.00
	144	5.00	1.00	2.00
	240	4.00	1.50	2.00
	480	8.00	2.00	2.00
10	96	12.00	1.00	1.00
	144	9.00	1.50	2.00
	240	2.50	2.00	2.00
	480	9.50	2.00	2.00

3.2 The Shrinkage Strategy: Priors vs. Data

This section compares hard lag selection against dynamic BVARs. We analyze both the bias-variance tradeoff and how the optimal prior relaxes asymptotically as sample sizes increase.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage (all fixed- τ BVAR-WN and BVAR-RW, plus MDD-selected τ).

T	p_0	AIC	BIC	HQC
96	4	1.681	1.312	1.085
	6	1.729	0.795	0.612
	8	1.457	1.108	0.444
	10	1.551	0.688	0.688
144	4	2.495	0.668	1.645
	6	0.810	0.514	0.514
	8	0.653	1.212	0.234
	10	0.211	2.019	1.700
240	4	0.890	1.310	1.310
	6	0.259	0.575	0.575
	8	1.263	0.993	1.472
	10	2.215	2.630	2.630
480	4	1.860	3.062	3.062
	6	1.000	0.546	0.546
	8	1.000	0.155	0.155
	10	1.000	0.252	0.252

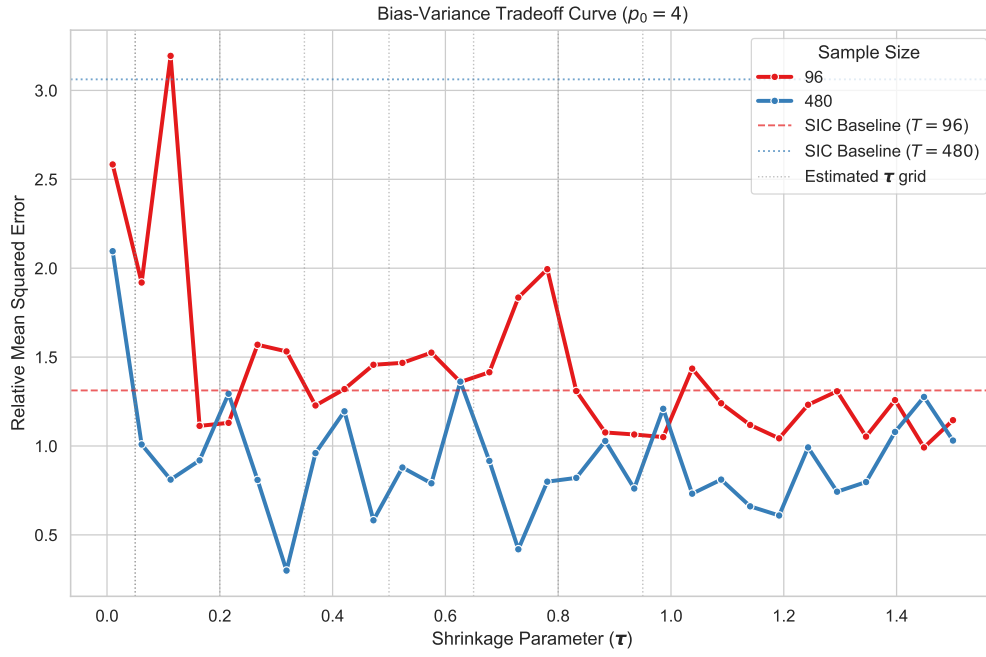


Figure 2: Comparison of Bias-Variance Tradeoff: BVAR vs. Hard Selection. The curve illustrates how the optimal τ balances bias and variance, achieving superior MSE performance compared to fixed lag selection.

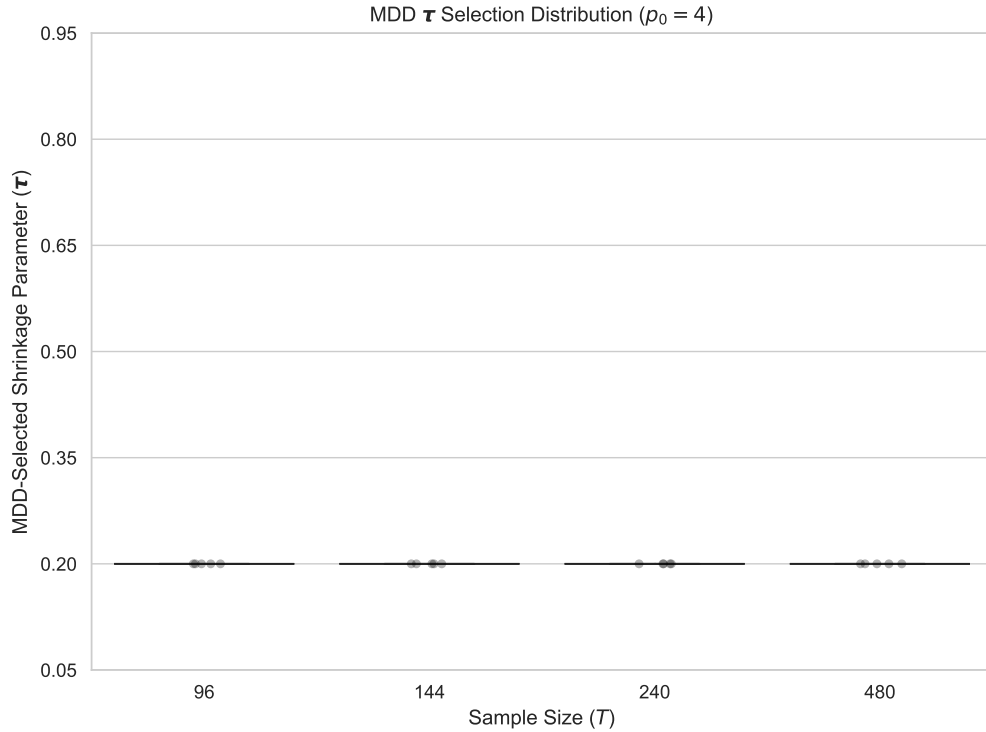


Figure 3: Asymptotic Relaxation of the Minnesota Prior. As the sample size (T) increases, the optimizer naturally selects looser priors, letting the data dominate the estimation.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.